Ethics and Information Technology (2025) 27:36
https://doi.org/10.1007/s10676-025-09845-2

ORIGINAL PAPER

ChatGPT is incredible (at being average)

Ihor Rudko[1] · **Aysan Bashirpour Bonab**[2]

Published online: 18 July 2025
© The Author(s) 2025

Abstract
In this article, we examine a peculiar issue apropos large language models (LLMs) and generative AI more broadly: the frequently overlooked phenomenon of output homogenization. It describes the tendency of chatbots to structure their outputs in a highly recognizable manner, which often amounts to the aggregation of verbal, visual, and narrative clichés, trivialities, truisms, predictable argumentations, and similar. We argue that the most appropriate conceptual lens through which said phenomenon can be framed is that of Frankfurtian bullshit. In this respect, existing attempts at applying the BS framework to LLMs are insufficient, as those are chiefly presented in opposition to the so-called algorithmic hallucinations. Here, we contend that further conceptual rupture from the original metaphor of Frankfurt (1986) is needed, distinguishing between the what-BS, which manifests in falsehoods and factual inconsistencies of LLMs, and the how-BS, which reifies in the dynamics of output homogenization. We also discuss how issues of algorithmic biases and model collapse can be framed as critical instances of the how-BS. The homogenization problem, then, is more significant than it initially appears, potentially exhibiting a powerful structuring effect on individuals, organizations, institutions, and society at large. We discuss this in the concluding section of the article.

Keywords ChatGPT · Large language model · Chatbot · AI slop · Homogenization · Bullshit

By this art you may contemplate the variation of the 23 letters.

—Anatomy of Melancholy, Pt. 2, Sec. II, Mem. IV, as quoted by Borges in The Library of Babel.

Introduction

The need to understand, anticipate, and prevent the adverse consequences of the widespread usage of generative AI is more pressing than ever (Hannigan et al., 2024; Markelius et al., 2024). A brief overview of the related literature confirms that attempts toward proactive recognition of chatbots' flaws are vast and growing. Research on algorithmic hallucination (Maleki et al., 2024; Salvagno et al., 2023; Smith et al., 2023) abounds. The same is true for the research on algorithmic biases (Gichoya et al., 2023; Rastogi et al., 2022; Roselli et al., 2019) and the risks of model collapse as related to synthetic data usage in model training (Dohmatob et al., 2024; Gerstgrasser et al., 2024; Shumailov et al., 2024). It is, therefore, surprising that the literature largely overlooks another fundamental dynamic with regard to generative AI: the tendency of chatbots to homogenize their outputs. So far, the topic has remained on the fringes of academia, with the exception of a few recently published papers focusing on the phenomenon of LLM output homogenization. For this precise reason, the bulk of our arguments will revolve around LLMs. Yet, the generalization is possible not only for the broader class of generative AI but also in relation to most instances of machine learning based on the connectionist paradigm (i.e., modern-day artificial neural networks).

In presenting our line of reasoning, we start by critically reviewing the idea of framing LLM outputs through the lens of Frankfurtian bullshit—i.e., when the interlocutor is indifferent to the truthfulness of their statements (Fisher, 2024; Frankfurt, 1986; Hicks et al., 2024). We argue that such metaphorical re-conceptualization is necessary but not

✉ Ihor Rudko
ihor.rudko@bi.no

✉ Aysan Bashirpour Bonab
aysan.bashirpourbonab@unicas.it

[1] BI Norwegian Business School, Oslo, Norway

[2] University of Cassino and Southern Lazio, Cassino, Italy

sufficient to understand how those machines work and to fully grasp the in-depth societal consequences of their further development and usage.

In the two recent papers in which bullshit re-conceptualization was proposed (Fisher, 2024; Hicks et al., 2024), it was explicitly presented as the alternative to a conventional "hallucination" framework. The latter portrays algorithmic falsehoods as troublesome occurrences that disrupt the otherwise normal operation of language models (Slater & Humphries, 2025). The bullshit re-conceptualization of Hicks et al. (2024) and Fisher (2024) measures itself against this very notion and, as a result, implicitly treats other problematic manifestations of chatbot's functionality as of lesser importance. This is an uncomfortable theoretical shortcoming to which we would like to propose a conceptual remedy.

By further elaborating on the bullshit metaphor, one necessarily risks deviating excessively from Frankfurt's (1986) original notion. We contend that this is a risk worth taking. In the first place, we don't know whether Frankfurt himself would be comfortable extending his concept to algorithmic entities such as chatbots. Besides, Hicks et al. (2024) already took some liberties with the original metaphor by arguing that LLMs are only "soft" kinds of bullshiters; that is, they satisfy some but not all original criteria of Frankfurtian bullshit (Fisher, 2024). While we are fully aware of the dangers of excessive nuance as far as good theorizing is concerned (Healy, 2017), in the present case, we argue it is not only unavoidable but necessary.

The authors advocating for the bullshit metaphor also argue that the emphasis on AI hallucinations excessively anthropomorphizes the chatbots (Slater & Humphries, 2025), ascribing to them human-like qualities and characteristics that are just not there (Fisher, 2024; Heersmink et al., 2024; Hicks et al., 2024). While we agree with the above argumentation, we believe that focusing on the apparent truthfulness of the chatbot's utterances is not sufficient. An excessive anthropomorphization of LLMs forecloses not only the superficial lexical-semantic edifice of the generated text but also the remainder of the complexity, nuances, and intricacies of written language. As such, it also forecloses some of its essential functions, which go beyond mere communication of ideas, events, results, etc. A similar observation is central to the prominent critique of the BS metaphor, mainly that in his original essay (Frankfurt, 1986) and then in the short book that followed it (Frankfurt, 2005), Frankfurt fails to acknowledge the dynamic and multifaceted nature of the very notion of truth (Perla & Carifio, 2007).

As Žižek succinctly puts it, language "should be twisted, denaturalized, extended, condensed, cut, and reunited, made to work against itself […] To extract the intended truth from it, [one] has to refer to literal meaning of words, to counter-intuitively displace the accent and scansion of the sentence, to translate single terms in a strongly-interpretative descriptive way" (Žižek, 2014: 565). In brief, the truthfulness of the written text resides not only in what it factually describes but also (and fundamentally) in how it is written, including the intricacies of the writing process itself. Paradoxically, simple declarative sentences, conciseness, apparent care for rigor, balanced exposition of different viewpoints, etc., do not always guarantee that text would be perceived as more valid or accurate. For instance, LLMs are surprisingly good at presenting information in a straightforward, polished, well-structured manner (Sastre et al., 2024). Is it then not surprising that experts in different fields consistently prefer texts written by other humans—be it the domain of legalistic enterprise, academic, or literary fiction (Cheng et al., 2023; Harasta et al., 2024; Marco et al., 2024)[1]? An obvious conclusion in line with Žižek's arguments is that there is something else, something different, enshrined not only in the plain lexicon but in the form itself, that sets human writing apart from that of even the most advanced chatbot. The main challenge is that this factor, or better, a collection of factors, is not easy to convey intuitively.

In this respect, the bullshit metaphor is still perfectly valid but needs to be partially de-coupled from its Frankfurtian foundations (Frankfurt, 1986). Accordingly, in the remainder of the paper, we argue for the usefulness of further distinguishing between two types of algorithmic bullshit: the what-BS and the how-BS. The first type broadly aligns with the initial conceptualization of Hicks et al. (2024) and Fisher (2024) and has mainly to do with the perceived factuality of the synthetic text. Hence, within our revised framework, a hallucination (i.e., algorithmic falsehood or even a complete word salad) is a critical manifestation of the what-BS. Conversely, the how-BS has to do with idiosyncrasies and nuances of the form itself. As much as models do not care about the truthfulness of their statements, they do not fundamentally care about how those statements are presented. The how-BS, as we will discuss, mainly manifests in the dynamics of output homogenization (Agarwal et al., 2024; Padmakumar & He, 2024). The latter results from the very logic of aggregation and optimization upon which today's complex connectionist AI relies (Endacott & Leonardi, 2024). Accordingly, like in the case of what-BS, the how-BS is not a model's flaw but is a result of its normal functionality.

[1] It is also worth noting that non-experts frequently prefer synthetic texts to human-written ones, like in the case of LLM-generated poetry (Porter & Machery, 2024). While the reason for this is unclear, it could relate to what Jonathan Gilmore, a philosophy professor at the CUNY, describes as the "banal, realistic style" of AI, which is "very easy to process" (Judkis, 2024). The observation is about the nature of synthetic images. Yet, one could make a similar argument in relation to AI-generated texts.

Our argumentation amounts to the following four propositions:

1. Chatbots cannot think. No matter how complex, they are "statistical brutes" that do not care about the nature of their outputs. The most appropriate metaphorical framework to describe the results of their normal functionality is that of Frankfurtian bullshit (Frankfurt, 1986; Hicks et al., 2024).
2. The conventional bullshit framework is necessary but not sufficient. First, it entirely subsumes the "hallucination" metaphor. The latter, we argue, is of great practical relevance and should not be rendered latent. Second, the framework ignores another fundamental property of LLMs: their proclivity for homogenizing outputs.
3. We propose distinguishing between the what-bullshit and the how-bullshit. The separation is mainly conceptual. The two types of bullshit are, indeed, an entangled duality of sorts (although, in their extreme manifestations, they can be antagonistic).
4. All an LLM can do is produce careless speech (Wachter et al., 2024). Hence, the two types of algorithmic bullshit are constantly at work. Reasoning in degrees of intensity is nonetheless useful. The what-BS manifests in the factual inconsistencies and falsehoods in the model's outputs. The how-BS reifies whenever the chatbot's homogenizing effects are clearly detectable and cumulatively relevant. In this regard, issues of algorithmic biases and model collapse can, in part, be framed as critical instances of output homogenization.

We will now proceed along these lines. The brief discussion of the societal implications of output homogenization concludes the paper.

ChatGPT is bullshit, but there is more to it

In June 2024, a conceptual article titled *ChatGPT is bullshit* (Hicks et al., 2024) went semi-viral on public forums, in particular, different technology and academia-related subreddits. When this paragraph was first written (the evening of February 9th, 2025), the paper had been cited 124 times on Google Scholar, an achievement that, in part, could be attributed to its brilliant title[2]. In the article, the authors argue for redefining how we think about LLM outputs, particularly in relation to their tendency to hallucinate.

Model hallucination is commonly understood as an occasional generation of content that "diverges from the user input, contradicts previously generated context, or misaligns with established world knowledge" (Zhang et al., 2023: p. 1). As an example, when we asked DeepSeek—the Chinese chatbot that in early 2025 took the market by surprise due to its open-source features, lower pre-training costs, and surprisingly good performance on several benchmarks (Gao et al., 2025)—who the author of the little epigraph to our paper is, it replied without hesitation:

The passage you're referencing is from Jorge Luis Borges' short story 'The Library of Babel' (1941).

In short, the model has hallucinated. The original quote, although famously used by Borges as an epigraph to his *The Library of Babel*, is borrowed from The Anatomy of Melancholy, a lengthy Renaissance treatise by Robert Burton, first published in 1621. The chatbot then went off tangents, seeking to provide a convincing justification for the number 23:

While the exact line in most translations refers to 25 letters (including spaces, commas, and periods), your version mentions 23, likely due to a misquotation or translation variation.

Not only has it got the author wrong, but it has also come up with a fictitious passage about misquotation (the truth is that the Classical Latin alphabet at the time consisted of 23 letters). ChatGPT was only slightly better at guessing the plausible source of the quote. At least, it was chronologically closer to the original text. The rest was again algorithmic fiction:

The quote is attributed to Johannes Trithemius, a German Benedictine abbot, cryptographer, and polymath. He wrote it in his work Steganographia (c. 1499), a treatise that appears to be about magic but is actually an early work on cryptography.

Many believe that the tendency of models to hallucinate is a substantial hurdle when it comes to ensuring their reliability in real-life applications (Bai et al., 2024; Chen et al., 2023; Jiang et al., 2024a, b; Liu et al., 2024a, b; Tonmoy et al., 2024; Wang et al., 2023). Some researchers even go so far as to suggest that hallucinations are inevitable and constitute an innate limitation of LLMs (Routray et al., 2023; Xu et al., 2024). Hicks et al. (2024), while sharing the general preoccupation regarding the proclivity of chatbots for generating falsehoods, consider the framing of false and contradictory outputs of LLMs as hallucinations fundamentally wrong. According to the authors, believing that models occasionally hallucinate erroneously suggests that when the models' answers sound legitimate, they are

[2] It is only fair to acknowledge that framing LLM outputs in terms of Frankfurtian bullshit has been also proposed by other scholars concurrently with Fisher and Hicks et al. (Gorrieri, 2024; Wachter et al., 2024) and before, in several periodicals (e.g., Bergstrom & Ogbunu, 2023; Marcus & Davis, 2020; McQuillan, 2023; Romero, 2023). Shannon Vallor has also described LLM outputs in terms of Frankfurtian BS in her recent book, *The AI Mirror: How to Reclaim Our Humanity in an Age of Machine Thinking* (Vallor, 2024). For a more detailed overview of the LLM BS literature, see Gunkel and Coghlan (2025).

not hallucinating. On the contrary, due to the ways in which LLMs function, all they can do is, in a way, "hallucinate." Sometimes, the generated statements are demonstrably false and misleading. Often, they are believable enough to pass for true. According to the authors (Hicks et al., 2024), because of emerging conceptual and practical problems associated with the term "hallucination," the better way to frame the overall output of LLMs is that of "bullshit," as famously formulated by the philosopher Harry Frankfurt (Frankfurt, 1986). "Frankfurt understands bullshit to be characterized not by an intent to deceive but instead by a reckless disregard for the truth" (Hicks et al., 2024: 4). Hence, when a so-called hallucination occurs, a better analogy for it would be that of a schoolchild not knowing the answer to a teacher's question but compelled to talk their way out of a bad mark.

The analogy, however, is not perfect if we were to apply the original concept of Frankfurt (Wachter et al., 2024). The problem as far as chatbots are concerned is that the analogy depicts a bullshiter with an intent. As such, it jointly satisfies the four essential criteria of Frankfurtian bullshit, as summarized by Fisher (2024):

1. The bullshitter is indifferent toward whether what they say is true or false.
2. The bullshitter is indifferent toward their audience's beliefs.
3. The bullshitter intends to deceive their audience into thinking that they are not bullshitting.
4. Bullshitting and lying are incompatible.

The mismatch between the original conceptualization of Frankfurt and that of algorithmic BS is discussed in depth by Wachter et al. (2024). The authors advance the alternative notion of "careless speech" to tackle the said mismatch (Wachter et al., 2024)[3]. The issue is also aptly recognized by Hicks et al. (2024), who argue that LLM bullshit is inherently limited to the first two propositions: the models are fundamentally indifferent toward the content of their speech and their users' beliefs. The third criterion is somewhat problematic when it comes to chatbots, as it inadvertently implies certain agentic qualities that contemporary AI systems simply do not possess (Vanneste & Puranam, 2024). The actual agents, one would argue, are the designers and the rest of the individuals responsible for the model's training. This includes RLHF (reinforcement learning from human feedback) workers who fine-tune the model to make its outputs more human-like (Wachter et al., 2024). In this respect, the intent on behalf of all those responsible for the chatbot's deployment to actively pursue the deception seems highly unlikely (Gunkel & Coghlan, 2025). Even in extreme instances when RLHF workers can provide only subjective perceptions of the truthfulness or accuracy of the model's outputs (Hawkins & Mittelstadt, 2023), making a case for the intent to deceive is plainly wrong.

We believe that the re-conceptualization of Hicks et al. (2024) is essential as far as chatbot ontology is concerned. While it is tempting to ascribe supreme agentic qualities to existing large language models (Vanneste & Puranam, 2024), those remain limited by (1) mathematical properties on which they rely (Barandiaran & Almendros, 2024) and (2) the agency of people/organizations involved in their creation and maintenance (Hicks et al., 2024). Ontology aside, the bullshit metaphor, as proposed by Hicks et al. (2024), is somewhat problematic from an epistemic but also practical standpoint (Gunkel & Coghlan, 2025) as it conceals the hallucination problem by merely restating, albeit by drawing on a different metaphor, the grim prospect of impossibility for LLMs to stop generating occasional falsehoods (Routray et al., 2023). Accordingly, by framing all LLM outputs as bullshit, we risk deemphasizing the importance of critical real-life examples of factually inconsistent synthetic texts, like when models generate false court citations (Milmo, 2023) or non-existing academic references (Gravel et al., 2023).

The metaphor also conceals further aspects of the chatbot's outputs that are equally relevant but have little to do with accuracy and truthfulness—at least from a superficial point of view. A more thorough perspective suggests that the question of truth runs deeper than the mere absence of algorithmic hallucinations. As we have briefly discussed in the Introduction, a factually impeccable synthetic text can still be problematic. Unfortunately, recent extensions of the bullshit metaphor to the language models have little to offer in this respect.

To grasp the nature of LLM outputs more accurately, we argue that an additional breakaway from the original bullshit conceptualization of Frankfurt (1986) is needed. To begin with, a fundamental distinction should be made between the factual consistency of synthetic text and how the latter is presented in terms of the heterogeneity of its linguistic features. As such, the distinction mainly opposes the content and the form, the "what" and the "how" of language, the basic meaning of the text, and the more latent and subtle ways in which the said meaning is conveyed through the corpus of features that could more proximally designate something akin to the chatbot's writing style, register or tone.

We propose to further distinguish between the what-bullshit and the how-bullshit of LLMs. From the conventional bullshit perspective (Fisher, 2024; Hicks et al., 2024), algorithmic hallucinations are not distinguishable from

[3] The problem with the "careless speech" framework is that, by definition, it best describes synthetic texts, whereas the BS framework can also apply to image, audio, and video generation.

factually correct statements as both are the result of the chatbot's bullshiting. The utility of preserving the distinction between algorithmic fabrication and truth is, nonetheless, substantial (Gunkel & Coghlan, 2025). An obvious solution to the dilemma is to consider hallucinations as particularly relevant and undesirable instances of the what-BS. In other words, the what-BS, although always at work, reifies in the form of algorithmic hallucination.

When it comes to the how-BS, algorithmic bullshiting mainly manifests in predictable and cliché ways in which chatbots structure their outputs beyond their superficial semantic signification. Namely, if a model consistently generates responses that are similar (in terms of their lexical and semantic content, style, tone, etc.) regardless of input variations, it homogenizes its outputs. Like hallucinations, the phenomenon of output homogenization is also the by-product of the normal functionality of LLMs. Yet, it is more insidious and difficult to spot as it is often harmless and inconsequential, especially on the level of individual prompts. Its cumulative effects are, however, of broader significance. For example, the issues of algorithmic biases and model collapse due to the reliance on synthetic data in training sets are intrinsically related to output homogenization. These, however, cannot be easily explained by means of conventional (what-) bullshit metaphor (Fisher, 2024; Hicks et al., 2024).

In the later section, we elaborate on the two types of algorithmic bullshit in more detail. Given that our conceptualization of what-BS is in line with that already well-described by Hicks et al. (2024) and Fisher (2024), we will mainly emphasize the how-BS and the ways it manifests in the phenomenon of output homogenization.

But first, a clarifying example might be helpful.

On glasses half-full of wine and other clichés

To better illustrate the nature of the how-BS and the related phenomenon of output homogenization, a visual example is perhaps the best place to start. Language, after all, is highly symbolic and is not conducive to straightforward intuitions. Notwithstanding, the exemplified dynamics are reasonably transferable to synthetic texts, as the technologies underlying heterogeneous modalities of generative AI are not so different at their conceptual core[4].

Let's start with the case where the boundaries between the what-BS and the how-BS are somewhat blurred. It turns out that until recently—before autoregressive image generation partially tackled this particular issue—it was almost impossible to make a chatbot generate a glass full of wine (Richardson, 2025). In most cases, no matter how specific the prompt was, the resulting image was only a half-full glass of wine (see, for example, the related discussion on Reddit: Accomplished_Tank184, 2024). A similar occurrence (still persistent) has been observed with pictures of analog clocks: no matter the hour specified by the user, the result is always a clock with the two hands pointing in the opposite direction in the two upper quadrants of the clockface, showing approximately the 10:10 (or 01:50).

Both examples are just more viral and known instances of manifest incoherence on behalf of chatbots. Yet, simply by understanding the likely cause of these abnormalities, one might come up with countless other examples of "impossible" prompts. Out of our head: generating an image of a person writing with the left hand, an image of a quarter of garlic clove, an image of a book opened on the very first page, an image of a T-Rex that is not tilted forward, an image of a tree without a bark (Fig. 1). It is also worth noting that while more famous of these instances might get "fixed" in the future—which, like in the case of wine glasses, was a rather successful marketing move by OpenAI (Richardson, 2025)—one can always think of additional examples of how to discombobulate the chatbots.

The likely explanation is somewhat unremarkable: people writing with their right hand, uncut cloves of garlic, and books opened roughly in the middle are overrepresented in the training set (Palmini & Cetinic, 2024). What we then observe is the model naturally gravitating to the most iconic or common representation, which is often nothing more than a mere cliché or, to put it even more blatantly, an algorithmic version of the "reversion to mediocrity[5]." Chatbot cannot simply depict a quarter of a clove of garlic or a book opened on the first page because it has not "seen" a sufficient amount of these in proportion to the majority of garlic cloves and opened books on the Internet. The frequently proposed solution to the problem is also unremarkable: more data, and especially more quality data (Wachter et al., 2024). However, even with all the data out there, how can one be sure that all the instances of all possible infinite prompts are sufficiently covered (Xu et al., 2024)? This is essentially impossible, and thinking that more data and/or better labeling could eventually help the model escape the "eigenvector trap" (Mukhopadhyay, 2025) is tantamount to putting out a fire with gasoline. On the contrary, one might easily overfit the model and exacerbate overrepresentation issues (Wolfe & Caliskan, 2021)[6].

The examples also show the intrinsically ambiguous and context-dependent nature of algorithmic hallucinations. If

[4] This is particularly true regarding recent advances in image generation. The so-called autoregressive (native) image generation is, all in all, a next-token prediction performed by LLM.

[5] Mukhopadhyay (2025) has recently advanced what is, in our view, the most apt term for the phenomenon: the "eigenvector trap."

[6] For the in-depth analytical intuition behind our reasoning, a curious reader can refer to Castro et al. (2023) and Xu et al. (2024).

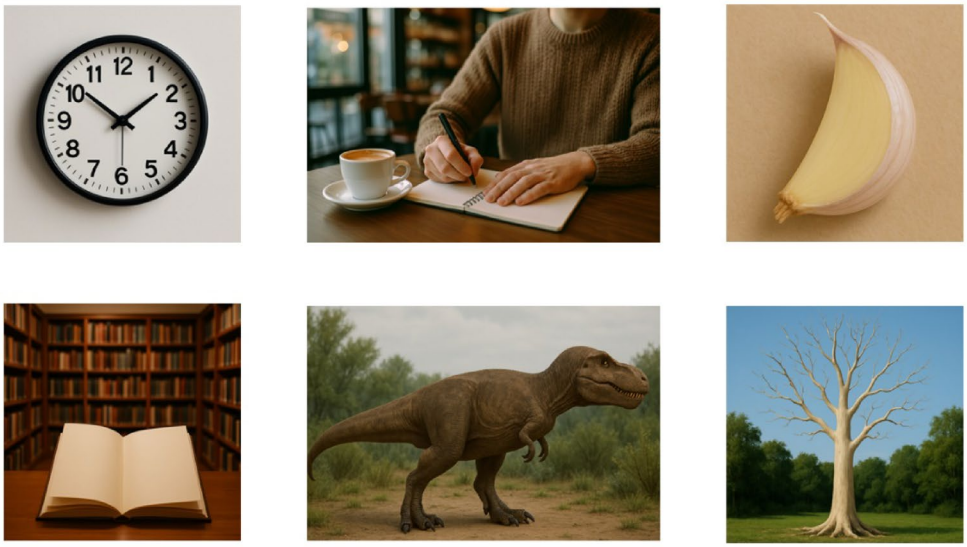

Fig. 1 Clockwise, from the top left: a clock showing 12:00, a person writing with the left hand, a quarter of garlic clove, a book opened on the very first page, a T-Rex that is not tilted forward, a tree without a bark. The images were generated natively by ChatGPT 4o. Note that we do not exclude that satisfactory results could still be reached with additional clever promoting

Fig. 2 Output homogenization at work (generated by ChatGPT 4o). The prompt was to create an image of a table full of glasses of red wine. The amount of liquid in each glass was not specified

we ask for an image of a glass full of wine and the result is a half-full glass instead, we might conclude that the model is hallucinating. But what if we prompt a glass of wine without further specification, and each time, the glass is half full? What if we ask for a table full of glasses of red wine, and all the glasses uncannily contain an identical amount of liquid (like in Fig. 2)? Is algorithmic hallucination still at work here? Probably not, at least not straightforwardly so, since in the last two cases the output is perfectly aligned with the prompt (if nothing else is specified). Is chatbot bullshiting in the sense of Hicks et al. (2024)? The answer is a resounding yes because, definitionally, all it can do is bullshit. Where the more conventional BS metaphor fails, however, is in capturing a subtle distinction between the form and manifest content. If we do not specifically prompt a half-full glass of wine, and the result is the glass of wine that is half-full, the output is, in the most superficial sense, non-confabulatory. Yet, this is only true until we start thinking more deeply about how the particular wine glass generated by the chatbot converges to an average representation of the wine glass that itself results from an overrepresentation of (half-full) glasses of wine in the training set of a model.

The examples given so far are mostly straightforward. However, when it comes to algorithmic bullshiting in image generation, one might get significantly more nuanced and specific. For instance, isn't it true that most AI-generated images are uniformly exposed in the background and the foreground and have a "plasticky" feeling of unnatural perfection about them (Rueda-Arango et al., 2024; Wasielewski, 2024)? Or that the individuals in portraits, unless explicitly prompted, are mostly located in the center of the frame (Ragot et al., 2020)? Or that the texture of human skin looks uniform, the anatomy is often distorted, the perspective is lacking, and there is no good distinction between fore-middle-background (Edwards, 2023)? Unsurprisingly, online communities were the first to notice and scrutinize the idiosyncrasies of synthetic images— hence, the rise in popularity of the term "AI-slop," which describes the inundation of the Internet by low-quality, low-effort AI-generated content (Hoel, 2024; Mahdawi, 2025; Sang et al., 2024). In this regard, the term "slop" has seen a 332% increase in usage in 2024 and has been shortlisted as Oxford

Word of the Year (*Oxford Word of the Year 2024*, 2024). Conversely, researchers have yet to follow suit[7].

The conceptual leap from image generation to text generation is not an easy one, as it requires a certain degree of abstraction. What is, after all, an equivalent of uniformly exposed background and foreground or a "plasticky" skin when it comes to synthetic text? In other words, what is the effect of the overrepresentation in training set in relation to not only the manifest semantic content of the text but, first and foremost, the more nuanced and subtle ways of linguistic structuring that can be tentatively described as "feel," "tone," "register," "writing style," or "manner of speaking" of the chatbot? The answer is not an easy one. After all, the famous McLuhanian dictum states that the medium is also the message (McLuhan, 1964). Likewise, the line between the content and the form of the chatbot's utterances is incredibly blurry. Nonetheless, we argue that distinguishing between the what-BS and the how-BS is still extremely useful. In the following section, we offer a more detailed explanation of why this is the case.

What-bullshit, how-bullshit, and their prime manifestations

In our re-conceptualization, the first subtype, the what-BS, is the easiest to define. As we have already argued, the original conceptualization of Hicks et al. (2024) is somewhat problematic. This is mainly due to the practical importance of distinguishing algorithmic hallucinations from factually impeccable outputs. What the conventional bullshit metaphor does to the hallucination framework is to subsume it entirely: both algorithmic falsehoods and a seemingly correct reply to a prompt are considered bullshiting of the same order of magnitude because, by definition, the model is (1) indifferent toward whether what it says is true or false and (2) is equally indifferent toward prompter's beliefs and expectations (Fisher, 2024).

If we want to keep the distinction between factually correct outputs and hallucinations intact, a proper way to define the what-BS is by directly relating it to the perceived factuality and coherence of the chatbot's outputs. Going back to the example of wine glasses, if by promoting the glass of wine, the chatbot generates, say, a hedgehog, the manifest what-BS is substantially more relevant than in the case of the model producing an actual image of a glass of wine. Identical consideration applies to the text: if the textual output of LLM is inconsistent in relation to the original prompt, either because the model "confuses" facts, names, or dates or merely because it goes off on a semantic tangent, then the what-BS is clearly significant and is not to be ignored.

What is important here is that the manifest what-BS should not be understood as the model making blatant mistakes purposefully but rather as a by-product of its normal functionality[8]. In this respect, Xu et al. (2024) provided formal proof of the inevitability of LLM hallucinations. In their own words: "Hallucination is inevitable for any computable LLM, regardless of model architecture, learning algorithms, prompting techniques, or training data" (Xu et al., 2024: 2). We have already presented an intuitive explanation for this predicament in the previous section of the article. In brief, the potentially infinite complexity of real-world scenarios is unmatched by LLMs' representational capabilities encoded in their semantic space (Xu et al., 2024). As a corollary, even if an answer to a given prompt is factually correct and/or semantically coherent, it is inevitable that, if further prompted, any given language model will eventually hallucinate.

The what-BS is also easier to conceptualize because the examples of LLMs generating falsehoods are numerous and often quite glaring. As already mentioned, while constantly at work, the what-BS primarily manifests as so-called algorithmic hallucinations. These are only innocuous if one merely plays around with the model. But what happens, for instance, if textual outputs produced by the chatbot are used for decision-making purposes in real-life scenarios when the non-negligible consequences for the user of the model but also other people involved are contingent on the veridicity of the text?

A salient group of examples pertains to the generation of fake court rulings within the common law legal systems. In June 2023, The Guardian reported that two US lawyers were fined because they had submitted fictitious court citations generated by ChatGPT (Milmo, 2023). A similar accident occurred at the time we were writing this paragraph: an Australian lawyer was caught using ChatGPT to reference nonexistent cases in filed court documents (Taylor, 2025). The instances of legalistic fiction generated by LLMs are not limited to ChatGPT. In December 2023, The Verge reported that former Trump lawyer Michael Cohen accidentally cited made-up course cases using Google's proprietary chatbot, Google Bard (currently rebranded as Gemini), allegedly by mistaking it for an advanced search engine (Roth, 2023).

Because the problem of hallucinations is so evidently out there, researchers actively seek measures to ensure LLMs' trustworthiness through sophisticated detection and prevention methods (Ji et al., 2023; Ye et al., 2023). What is less evident is the "how" aspect of LLM bullshit—in other

[7] A following *in-title* query on Scopus on 15/02/2025 returned zero results:

TITLE ("AI" OR "LLM" OR "machine learning") AND TITLE ("slop").

[8] We do not intend to explain in detail how an LLM works. For further clarifications, one could refer to Briganti (2024) or alternatively to Hicks et al. (2024) for a more concise expanation.

words, different ways in which language models structure their outputs. The how-BS is inherently more subtle and escapes immediate categorization. Yet, it is by no means a residual category in the sense that we should only focus on the how-BS when the what-BS is reasonably negligible. The best way to think about the two is in terms of their relational entanglement—as inseparable and interdependent parts of a greater duality[9]. As far as our taxonomy is concerned, the how-BS manifests in rather predictable ways in which generated words relate to other words, sentences relate to sentences, paragraphs to paragraphs, and thematic clusters to other thematic clusters. When we assert that, independently of the specific prompt, a chatbot's way of writing has a particular style, feel, tone, register, mood, manner, etc… to it, we implicitly invoke the how-BS argument.

Returning to the example of wine glasses, one can think of the how-BS as a linguistic equivalent of always having a glass that is half full when this is not required or proscribed explicitly (see Fig. 2), but also of heterogenous defining characteristics of "AI art," such as homogenous and repetitive patterns, excessive symmetry, lack of imperfections, "smoothness," uniformity of color and composition, and so on (Edwards, 2023; Rueda-Arango et al., 2024; Wasielewski, 2024). Like in the case of what-BS, the how-BS is essentially negligible when the prompter is indifferent to its structuring consequences on the text and when those structuring consequences do not readily reveal themselves in the first place. Similarly to the what-BS, it is also always at work, as all LLMs can do, is engage in perpetual Frankfurtian bullshiting (Hicks et al., 2024), of which the how-BS is a constitutive part.

Just like the what-BS revealing itself through algorithmic hallucinations, the how-BS primarily manifests in the phenomenon of output homogenization. The latter, we argue, should be considered by AI researchers more scrupulously. This is not to say that no research has been done to understand whether LLMs exhibit a particular writing style (AlAfnan & MohdZuki, 2023), how synthetic text differs from the way humans write (Herbold et al., 2023), and how output homogenization might influence specific institutional fields (Geng & Trotta, 2024; Okerlund et al., 2022). Compared to the research on the mitigation of hallucinations, the scholarship on the homogenizing effects of LLMs is, however, relatively scarce. Or, better, related topics are emphasized instead, including different types of algorithmic biases (Reif et al., 2023) and the issue of model collapse (i.e., when models are trained on their own data, leading to the gradual degradation of outputs over time, including their substantial homogeny). While this emphasis is entirely justified due to the pressing importance of respective research, we believe that output homogenization requires more dedicated scholarly attention, as it is closely connected to but does not exhaust itself in algorithmic biases or model collapse. We will return to this point in the following section of the article.

By systematically searching for articles on the homogenizing effects of LLMs on Scopus and Web of Science and then by complementing the search with a number of in-depth Google Scholar queries, we were able to identify only 56 scholarly documents on the topic. Moreover, upon closer reading, we found that most of the identified documents were not pertinent to the subject matter or only mentioned it indirectly. What follows is a brief recap of the most important findings.

To begin with, homogenization is not uniquely relevant to LLMs or generative AI more broadly. The literature on the homogeneous outcomes of machine learning, a category that includes both, has been steadily growing over the last few years (Ajunwa, 2016; Creel & Hellman, 2021; Fishman & Hancox-Li, 2022; Jain et al., 2024; Wang & Russakovsky, 2023). As the respective research area is yet to be fully established, in all the cited instances, the authors speak of homogenization in different terms and through heterogeneous conceptual lenses—predominantly those of algorithmic biases (Shahbazi et al., 2023; Thrampoulidis, 2024).

When it comes to explicitly mentioning homogenization within the broader machine learning context, this has been done by only a handful of authors. Three papers are worthy of attention in this regard. Bommasani et al. (2022) introduced the concept of algorithmic monoculture. The latter comes into being when algorithmic systems share similar or identical components (like training sets or models) and are then used by multiple decision-makers. The risk, of course, is that homogenizing effects could propagate within and across organizations, industries, and institutions. Likewise, Toups et al. (2023) looked at homogenization at the ecosystem level of analysis. They found that across three modalities (text, images, speech) and eleven datasets, machine learning was prone to systemic failure when all the models used in the study misclassified some individuals at the expense of others. Finally, Endacott and Leonardi (2024) offered a more nuanced theoretical explanation of why homogenization happens. In their words, "algorithmic isomorphism occurs as AI technologies implement patterns

[9] It is also legitimate to argue that, when manifested more intensly, the what-BS and the how-BS are actually antagonistic and take the form of a conceptual dualism. For example, a conventional way of tackling the homogenization problem is to increase the model's temperature hyperparameter (higher temperature ensures that the model will select less probable next tokens in the respective distribution). However, this can also result in the increase of what-BS-related factual inconsistencies generated by the model. As noted by Sui et al. (2024:1), "the tendency for LLMs to confabulate may be intimately associated with a positive capacity for coherent narrative-text generation."

optimized for specific outcomes as gleaned from aggregated data across time and space" (Endacott & Leonardi, 2024: p. 343). Broadly speaking, the core of what all existing AI systems do is advanced mathematical optimization (McQuillan, 2022). The outcome of the said optimization is, then, closely determined by the sheer size of the aggregated data in the training set (Endacott & Leonardi, 2024). Therefore, it is only natural that machine learning outcomes are highly prone to homogeny. "AI technologies make an emergent form of isomorphism possible through the aggregation of human action, as optimized for a determined set of constraints" (Endacott & Leonardi, 2024: p. 343). The mere act of choosing a few relevant constraints from the literally infinite "menu" of options available in the real world ensures that some outcomes will never be prioritized or even considered. It does not matter much who chooses the said constraints: the user, the designer, or both.

The literature on LLM output homogenization is somewhat richer and more empirically driven. According to Padmakumar and He (2024), LLMs tend to increase homogeneity when used in collaborative writing. They also tend to impose Americanized cultural-linguistic representations on non-English speakers by suggesting Western-centric autocompletes (Agarwal et al., 2024). When prompted to produce scientific reports, ChatGPT predominately references the most cited articles within the respective research area (Petiska, 2023). This, in turn, can favor hegemonic forms of knowledge by silencing the already underrepresented voices in academia, including scholars from the Global South (Pragya, 2024). In general, English is used as an implicit pivot language in most LLMs due to the sheer amount of respective content in training sets (Wendler et al., 2024). As a result, common patterns and structures of English tend to "migrate" to the output in other languages. ChatGPT-generated abstracts are consistently rated as subpar by human experts, and the resulting texts show lower burstiness and perplexity, which suggests their greater homogeneity (Cheng et al., 2023). LLMs tend to exhibit a recognizable writing style (AlAfnan & MohdZuki, 2023). The latter is marked by avoidance of passive sentences and overreliance on "difficult" words (Georgiou, 2024), less diversity in verb usage, and more analytical language (Alvero et al., 2024). Texts written with the assistance of chatbots are more similar to each other than those exclusively written by humans (Anderson et al., 2024; Doshi & Hauser, 2024). These findings hold even when a range of diversity-enhancement techniques is used to enrich the models' outputs (Moon et al., 2024). LLM responses are also frequently marked with excessive positivity and sycophancy (Wu et al., 2024; Zhang et al., 2024). More generally, models' outputs are more rigid and formulaic than human-written texts (Jiang & Hyland, 2024b) and tend to display a narrower worldview (Shur-Ofry et al., 2024). The content they generate often feels mainstream and standardized (Runco, 2025; Shur-Ofry, 2023). In this regard, LLMs are no match for world-class fiction authors (Marco et al., 2024). The homogenizing effects of LLMs might also have serious cognitive repercussions on users: the overreliance on chatbots in creative tasks results in the similarity of content, which carries on even when the model's usage is discontinued (Q. Liu et al., 2024a, b). Finally, one of the most comprehensive quantitative evaluations of LLM output homogenization to date (Sourati et al., 2025) highlights that, from 2023 and onward, there has been a substantial increase in LLM usage online, which coincides with a decrease in the diversity of texts as measured by a composite writing complexity index. The homogenization is also accompanied by systematic erasure of linguistic cues that convey unique personality profiles of human writers (Sourati et al., 2025).

Output homogenization, model collapse, algorithmic biases, and why those are related

The marginal role that AI researchers designated to output homogenization was somewhat surprising to us, especially in light of the growing relevance of adjacent research on algorithmic biases and model collapse. These, we argue, are closely related to the homogenization phenomenon and, consequently, to the dynamics of the how-BS. Let's look first at the issue of model collapse.

As a recent *Nature* article shows, training each successive generation of a language model on synthetic data produced by the previous generations has dire cumulative effects over time (Shumailov et al., 2024). After a few generations, researchers observed a massive degradation of the synthetic text. Interestingly, they notice that "data generated by language models in […] experiments end up containing a large number of repeating phrases[10]" (Shumailov et al., 2024: p. 758). Yet, the authors promptly dismissed output homogenization as of secondary importance. In their own words, "To rule this out as the cause of model collapse, we further provide numerical experiments when models are explicitly encouraged to produce non-repeating sequences" (Shumailov et al., 2024: 758). The authors, therefore, repeated the training process to find that, even when repetition was explicitly penalized, the model still slipped into gibberish. The obvious criticism here is that the penalty in question exclusively affected the repetition of lexical occurrences. Yet, the notion of the how-BS is important insofar as written language is more intricate and complex than its apparent

[10] An example from the article: "architecture. In addition to being home to some of the world's largest populations of black @-@ tailed jackrabbits, white @-@ tailed jackrabbits, blue @-@ tailed jackrabbits, red @-@ tailed wujackrabbits, yellow @-."

lexical-semantic edifice (Reviriego et al., 2024). To quote brilliant Elfriede Jelinek, "Language should be 'tortured' to tell the truth" (Jelinek, 1995: 72). The fundamental question then is what constitutes the said "torture." In this regard, we contend that the lexicon and surface-level semantics are only a small part of what a language model can theoretically homogenize[11].

The comment on the original article does a better job of emphasizing the role of the how-BS in successive models trained on synthetic corpora (Wenger, 2024). In the words of the author: "This model collapse occurs because training models on their own generated content causes them to 'forget' the less-common elements of their original training data set" (Wenger, 2024: 743). What is it, then, if not an implicit invocation of the output homogenization? The author continues by providing an example of a generative AI tasked with creating images of dogs and how those would eventually converge to the image of the most representative breed in the training set. This, of course, is a toy example. In terms of real-life consequences of LLMs, an important question to answer is: precisely because the how-BS is more "insidious" and its homogenizing effects often manifest in latent and subtle ways, what are the long-term implications of idiosyncratic, Wikipedia-esque (Le Bronnec et al., 2024), frequently plain, straightforward (yet, paradoxically, often described as verbose), machine-like way of writing inundating our communicative spaces?

One must first ask how serious the problem is—that is, how much synthetic text is already out there in the wild. The answer to this question is not simple. The so-called AI detectors are notoriously faulty and ridden with false positives and false negatives (Gritsai et al., 2025; Wu & Flanagan, 2023). Consequently, estimating how much content online, in percentage terms, is now generated by LLMs is somewhat of a guesswork. Yet, some estimations have already been given. A study from a team of Amazon Web Services researchers, for example, showed that up to 57% of all Web content could be generated by machines (Thompson et al., 2024). The study was submitted on arXiv in January 2024, so the current percentage is likely higher. What are, then, the consequences of such uncanny proliferation not only for the quality of future LLMs but also for society that increasingly relies on chatbots? As Žižek puts it in his recent book, "The true danger is not that we will misperceive what a chatbot is saying as the talk of a 'real' person; it is much more that communicating with chatbots will make 'real' persons talk like chatbots—missing the nuances and ironies, obsessed with false precision of saying only what they really want to say, etc." (Žižek, 2024: p. 178)[12].

Algorithmic biases are also related to the homogenizing effects of the how-BS (Castro et al., 2023). Yet, the two research areas overlap only to a certain extent; that is, not all biases are rooted in output homogenization, and, vice versa, not all output homogenization outcomes can be framed as algorithmic biases *in stricto sensu*. In this respect, the two research areas are complementary.

An example of an interesting emergent area of overlap is the so-called homogeneity bias, first introduced by Lee et al. (2024) and subsequently analyzed in detail by the same author and collaborators (Lee, 2025; Lee & Jeon, 2025; Lee & Lai, 2024). The homogeneity bias happens when a chatbot portrays particular ethnic, gender, religious, and other minority groups as more homogeneous than others (Lee & Jeon, 2025). A careful reader should, by now, intuit how the broader homogenization dynamics of the how-BS play out in this specific instance: marginalized groups, less populous nationalities, and other minorities are, by definition, underrepresented in training sets. Moreover, their very presence online is often contingent upon the salient features of ethnicity, nationality, gender, or religion they belong to. That is, as far as my online presence is concerned, I am a member of this or that group insofar as I stereotypically behave, dress, or talk like other group members[13].

In this regard, the most straightforward and obvious mitigation strategy is adjusting the model's hyperparameters (Attanasio et al., 2022; Zayed et al., 2024), first and foremost, the already-mentioned temperature (see footnote 9). In essence, raising the model's temperature increases the likelihood that the words/tokens that are different from the most probable will be chosen to complete the output sequence. (So instead of the proverbial elephant in the… room, we might as well obtain an elephant in the… space shuttle or an elephant in the… movie theater). By making

[11] The deeper connection between homogenizing tendencies of LLMs and the model collapse is discussed by Wu et al. (2024) in relation to the concept of generative monoculture. Likewise, Guo et al. (2024) observed a decline in output diversity (which they termed "curious") when successive generations of a model were trained on synthetic output produced by previous generations.

[12] Similar preoccupations have been recently put forward by scholars working within narrower academic fields, like that of law and governance. For instance, "Taking a close look at the technological underpinnings of LLMs, the analysis suggests that, due to the combination of human judgements, training datasets, and inherent features of the underlying technological paradigm—LLM outputs are likely to be geared toward the popular and mirror to their users concentrated, mainstream worldviews, sidelining a broader spectrum of perspectives" (Shur-Ofry, 2023:1).

[13] One can easily check this with minimal effort by simply searching "Americans" and then, for example, "Estonians" on Google Images. The first search returns a cast of individuals of different ages, skin tones, wearing more casual and varied clothes, while the second search converges on mostly young people (women, in particular) wearing predominantly ethnic clothing. Similar observation can be made apropos any other less populous nationality and/or ethnicity in relation to a more populous one.

the output more lexically unpredictable, one might hope to steer the model away from dominant patterns in the training data and to successfully avoid the "eigenvector trap." Unsurprisingly, and in line with our observation on the nuanced and multifaceted nature of output homogenization, adjusting the temperature (as well as other relevant hyperparameters), while somewhat effective in specific instances, is not a universal solution to homogeneity bias (Lee, 2025).

A couple of critical remarks on hyperparameters, output homogenization, and bias mitigation. First, the how-BS arguments we have advanced here run counter to the idea that hyperparameter tuning alone may effectively ensure that the model sounds less robotic and produces a more creative and engaging output (Davis et al., 2024; Nguyen et al., 2024). For instance, in their study of the temperature's effect on the quality of the generated texts, Peeperkorn et al. (2024: p. 1) conclude that "the influence of temperature on creativity is far more nuanced and weak than suggested by the 'creativity parameter' claim." Second, the actual risk of setting the temperature to the maximum is to increase the degree of semantic incoherence and, therefore, to make the what-BS more noticeable without necessarily reducing the manifest how-BS. Hence, the basic semantic consistency of the generated text effectively bounds the temperature range (Abdurahman et al., 2024). Third, the belief that algorithmic biases can be sufficiently mitigated by "tweaking" the model's hyperparameters forecloses the fundamental problem at the conceptual core of the how-BS metaphor: language models, especially their "large" variety, are essentially simplified statistical representations of our existing symbolic order. As such, they will always tend to reproduce and amplify biases and prejudices already present in the real world. Accordingly, keeping humans directly or indirectly "in the loop" when it comes to effectively de-biasing LLMs and aligning them with social values is as critical as ever.

Conclusion

The relatively little attention that output homogenization has received from scholars might seem puzzling at first. In reality, research on AI biases (Lee & Jeon, 2025) already partially addresses the phenomenon—albeit indirectly. The case is trickier for the issue of model collapse, i.e., when training the model on synthetic data leads to utterly broken chatbots (Shumailov et al., 2024). Regardless, one could come up with a convincing argument that homogenization is pretty much central here (Wenger, 2024).

A particular reason why output homogenization is per se so relatively underresearched is that as far as the further legitimation of chatbots is concerned (Grimmelikhuijsen & Meijer, 2022), issues of AI biases and model collapse are as critical to be addressed as chatbots' confabulatory tendencies (Grimmelikhuijsen & Meijer, 2022). Conversely, homogenization, as the phenomenon that partially overlaps but is nonetheless distinct from algorithmic biases and model collapse, is often innocuous or even a net positive. For example, it is always desirable for a diagnostic AI tool to detect a particular skin lesion as cancer in case it is actually one. Homogenization of (trustworthy) outputs is here strictly necessary and is merely a testament to the algorithm's efficacy. Yet, when the model's use cases increase, homogenizing effects are less straightforwardly positive. In the case of LLMs, these also go beyond algorithmic biases and model degradation due to synthetic training data. Think about the proliferation of "AI-slop" online and its effects on the future of the Internet (Mahdawi, 2025). Think about the potential harm to human creative endeavors when, for instance, screenwriting, design, animation, etc., are outsourced to chatbots just because they produce seemingly satisfactory results for a fraction of the costs (Rudko et al., 2024). To go even further, think about Žižekian predicament of human language increasingly resembling that of chatbots—"missing the nuances and ironies, obsessed with false precision of saying only what they really 'want to say'" (Žižek, 2024: p. 178). The most recent empirical findings suggest that this might be already happening (Lee et al., 2025).

Chatbots are frequently marketed as precursors to artificial general intelligence (Markelius et al., 2024; Routray et al., 2023). To prove the legitimacy of the claim, we are often presented with spurious benchmarks of chatbots' "intelligence" (Galleta, 2024; Wachter et al., 2024), proving that models can increasingly outperform humans on complex non-routine tasks. And, indeed, often they can. But what is it, if not sheer formal rationality reasoning in a strictly Weberian sense? The same rationality that, unlike its substantive counterpart, disregards the fundamental ethical considerations of our goals and is merely obsessed with the means of achieving them (Kalberg, 1980). The same rationality behind the major atrocities of the 20th century (Bauman, 2000). In a world where reactionary politics are on the rise, and the discourses of those in power are frequently centered on AI as an instrument of maximizing efficiency, exerting control, transcending human "irrationality," tempering our "flaws" and biases, and many similar (McQuillan, 2022), these are some good questions to ask oneself. The how-BS framework and the related phenomenon of output homogenization we have discussed here might seem highly abstract. They are, nonetheless, more relevant than ever. Chatbots are here to stay. In the end, what we call for is greater reflexivity in using them.

Acknowledgements The authors are grateful to Liam Rudko for the impetus and inspiration.The authors would also like to thank the two

anonymous reviewers for their insightful suggestions and comments.

Author contributions I.R. Conceptualization, writing, editingA.B.B. Writing, editing.

Funding Open access funding provided by Norwegian Business School. This study was not supported by any sponsor or funder.

Data availability No datasets were generated or analysed during the current study.

Declarations

Competing interests The authors declare no competing interests.

Open Access This article is licensed under a Creative Commons Attribution 4.0 International License, which permits use, sharing, adaptation, distribution and reproduction in any medium or format, as long as you give appropriate credit to the original author(s) and the source, provide a link to the Creative Commons licence, and indicate if changes were made. The images or other third party material in this article are included in the article's Creative Commons licence, unless indicated otherwise in a credit line to the material. If material is not included in the article's Creative Commons licence and your intended use is not permitted by statutory regulation or exceeds the permitted use, you will need to obtain permission directly from the copyright holder. To view a copy of this licence, visit http://creativecommons.org/licenses/by/4.0/.

References

Abdurahman, S., Atari, M., Karimi-Malekabadi, F., Xue, M. J., Trager, J., Park, P. S., Golazizian, P., Omrani, A., & Dehghani, M. (2024). Perils and opportunities in using large Language models in psychological research. *PNAS Nexus*, *3*(7), pgae245. https://doi.org/10.1093/pnasnexus/pgae245

Accomplished_Tank184 (2024, November 22). *Anyone know why it cannot do this?* [Reddit Post]. R/OpenAI. www.reddit.com/r/OpenAI/comments/1gxgns9/anyone_know_why_it_cannot_do_this/

Agarwal, D., Naaman, M., & Vashistha, A. (2024). *AI Suggestions Homogenize Writing Toward Western Styles and Diminish Cultural Nuances* (No. arXiv:2409.11360). arXiv. https://doi.org/10.48550/arXiv.2409.11360

Ajunwa, I. (2016). The paradox of automation as Anti-Bias intervention (SSRN scholarly paper 2746078). *Social Science Research Network*. https://doi.org/10.2139/ssrn.2746078

AlAfnan, M. A., & MohdZuki, S. F. (2023). Do artificial intelligence chatbots have a writing style?? An investigation into the style?istic features of ChatGPT-4. *Journal of Artificial Intelligence and Technology*, *3*(3). https://doi.org/10.37965/jait.2023.0267

Alvero, A. J., Lee, J., Regla-Vargas, A., Kizilcec, R. F., Joachims, T., & Antonio, A. L. (2024). Large Language models, social demography, and hegemony: Comparing authorship in human and synthetic text. *Journal of Big Data*, *11*(1), 138. https://doi.org/10.1186/s40537-024-00986-7

Anderson, B. R., Shah, J. H., & Kreminski, M. (2024). Homogenization Effects of Large Language Models on Human Creative Ideation. *Proceedings of the 16th Conference on Creativity & Cognition*, 413–425. https://doi.org/10.1145/3635636.3656204

Attanasio, G., Nozza, D., Hovy, D., & Baralis, E. (2022). *Entropy-based Attention Regularization Frees Unintended Bias Mitigation from Lists* (No. arXiv:2203.09192). arXiv. https://doi.org/10.48550/arXiv.2203.09192

Bai, Z., Wang, P., Xiao, T., He, T., Han, Z., Zhang, Z., & Shou, M. Z. (2024). *Hallucination of Multimodal Large Language Models: A Survey* (No. arXiv:2404.18930). arXiv. https://doi.org/10.48550/arXiv.2404.18930

Barandiaran, X. E., & Almendros, L. S. (2024). Transforming agency. *On the Mode of Existence of Large Language Models*. https://doi.org/10.48550/arXiv.2407.10735.

Bauman, Z. (2000). *Modernity and the holocaust*. Cornell University Press.

Bergstrom, C. T., & Ogbunu, C. B. (2023, April 6). ChatGPT Isn't 'Hallucinating.' It's Bullshitting. Undark Magazine. https://undark.org/2023/04/06/chatgpt-isnt-hallucinating-its-bullshitting/

Bommasani, R., Creel, K. A., Kumar, A., Jurafsky, D., & Liang, P. S. (2022). Picking on the same person: Does algorithmic monoculture lead to outcome homogenization?? *Advances in Neural Information Processing Systems*, *35*, 3663–3678. https://doi.org/10.48550/arXiv.2211.13972

Briganti, G. (2024). How ChatGPT works: A mini review. *European Archives of Oto-Rhino-Laryngology*, *281*(3), 1565–1569. https://doi.org/10.1007/s00405-023-08337-7

Castro, F., Gao, J., & Martin, S. (2023). *Human-AI Interactions and Societal Pitfalls* (No. arXiv:2309.10448). arXiv. https://doi.org/10.48550/arXiv.2309.10448

Chen, Y., Fu, Q., Yuan, Y., Wen, Z., Fan, G., Liu, D., Zhang, D., Li, Z., & Xiao, Y. (2023). Hallucination Detection: Robustly Discerning Reliable Answers in Large Language Models. *Proceedings of the 32nd ACM International Conference on Information and Knowledge Management*, 245–255. https://doi.org/10.1145/3583780.3614905

Cheng, S., Tsai, S., Bai, Y., Ko, C., Hsu, C., Yang, F., Tsai, C., Tu, Y., Yang, S., Tseng, P., Hsu, T., Liang, C., & Su, K. (2023). Comparisons of quality, correctness, and similarity between ChatGPT-Generated and Human-Written abstracts for basic research: Cross-Sectional study. *JOURNAL OF MEDICAL INTERNET RESEARCH*, 25. https://doi.org/10.2196/51229

Creel, K., & Hellman, D. (2021). *The Algorithmic Leviathan: Arbitrariness, Fairness, and Opportunity in Algorithmic Decision Making Systems* (SSRN Scholarly Paper No. 3786377). Social Science Research Network. https://papers.ssrn.com/abstract=3786377

Davis, J., Bulck, L. V., Durieux, B. N., & Lindvall, C. (2024). The temperature feature of chatgpt: Modifying creativity for clinical research. *JMIR Human Factors*, *11*(1), e53559. https://doi.org/10.2196/53559

Dohmatob, E., Feng, Y., Subramonian, A., & Kempe, J. (2024). *Strong Model Collapse* (No. arXiv:2410.04840). arXiv. https://doi.org/10.48550/arXiv.2410.04840

Doshi, A. R., & Hauser, O. P. (2024). Generative AI enhances individual creativity but reduces the collective diversity of novel content. *Science Advances*, *10*(28). https://doi.org/10.1126/sciadv.adn5290. Scopus.

Edwards, K. (2023, July 5). Why Does All AI Art Look Like That? *Medium*. https://medium.com/@keithkisser/why-does-all-ai-art-look-like-that-f74e2a9e1c87

Endacott, C. G., & Leonardi, P. M. (2024). Chapter 19*: Artificial intelligence as a mechanism of algorithmic isomorphism*. https://www.elgaronline.com/edcollchap/book/9781803926216/book-part-9781803926216-29.xml

Fisher, S. A. (2024). Large Language models and their big bullshit potential. *Ethics and Information Technology*, *26*(4), 67. https://doi.org/10.1007/s10676-024-09802-5

Fishman, N., & Hancox-Li, L. (2022). Should attention be all we need? The epistemic and ethical implications of unification in machine learning. *Proceedings of the 2022 ACM Conference on Fairness,*

Accountability, and Transparency, 1516–1527. https://doi.org/10.1145/3531146.3533206
Frankfurt, H. (1986). *On bullshit*. Princeton University Press.
Frankfurt, H. (2005). *On bullshit*. Princeton University Press.
Galleta, C. (2024, July 30). Big Techs Are Lying About LLMs Benchmarks… Again. *Medium*. https://medium.com/@thcookieh/big-techs-are-lying-about-llms-benchmarks-again-ef4e041d6396
Gao, T., Jin, J., T Ke, Z., & Moryoussef, G. (2025). A comparison of deepseek and other LLMs. *ArXiv*. https://doi.org/10.48550/ArXiv.2502.03688. No. ArXiv:2502.03688.
Geng, M., & Trotta, R. (2024). *Is ChatGPT Transforming Academics' Writing Style?* (No. arXiv:2404.08627). arXiv. https://doi.org/10.48550/arXiv.2404.08627
Georgiou, G. P. (2024). *Differentiating between human-written and AI-generated texts using linguistic features automatically extracted from an online computational tool* (No. arXiv:2407.03646). arXiv. https://doi.org/10.48550/arXiv.2407.03646
Gerstgrasser, M., Schaeffer, R., Dey, A., Rafailov, R., Sleight, H., Hughes, J., Korbak, T., Agrawal, R., Pai, D., Gromov, A., Roberts, D. A., Yang, D., Donoho, D. L., & Koyejo, S. (2024). *Is Model Collapse Inevitable? Breaking the Curse of Recursion by Accumulating Real and Synthetic Data* (No. arXiv:2404.01413). arXiv. https://doi.org/10.48550/arXiv.2404.01413
Gichoya, J. W., Thomas, K., Celi, L. A., Safdar, N., Banerjee, I., Banja, J. D., Seyyed-Kalantari, L., Trivedi, H., & Purkayastha, S. (2023). AI pitfalls and what not to do: Mitigating bias in AI. *British Journal of Radiology*, *96*(1150), 20230023. https://doi.org/10.1259/bjr.20230023
Gorrieri, L. (2024). Is ChatGPT full of bullshit?? *Journal of Ethics and Emerging Technologies*, *34*(1). https://doi.org/10.55613/jeet.v34i1.149
Gravel, J., D'Amours-Gravel, M., & Osmanlliu, E. (2023). Learning to fake it: Limited responses and fabricated references provided by ChatGPT for medical questions. *Mayo Clinic Proceedings: Digital Health*, *1*(3), 226–234. https://doi.org/10.1016/j.mcpdig.2023.05.004
Grimmelikhuijsen, S., & Meijer, A. (2022). Legitimacy of algorithmic Decision-Making: Six threats and the need for a calibrated institutional response. *Perspectives on Public Management and Governance*, *5*(3), 232–242. https://doi.org/10.1093/ppmgov/gvac008
Gritsai, G., Voznyuk, A., Grabovoy, A., & Chekhovich, Y. (2025). *Are AI Detectors Good Enough? A Survey on Quality of Datasets With Machine-Generated Texts* (No. arXiv:2410.14677). arXiv. https://doi.org/10.48550/arXiv.2410.14677
Gunkel, D., & Coghlan, S. (2025). Cut the crap: A critical response to ChatGPT is bullshit. *Ethics and Information Technology*, *27*(2), 23. https://doi.org/10.1007/s10676-025-09828-3
Guo, Y., Shang, G., Vazirgiannis, M., & Clavel, C. (2024). *The Curious Decline of Linguistic Diversity: Training Language Models on Synthetic Text* (No. arXiv:2311.09807). arXiv. https://doi.org/10.48550/arXiv.2311.09807
Hannigan, T. R., McCarthy, I. P., & Spicer, A. (2024). Beware of botshit: How to manage the epistemic risks of generative chatbots. *Business Horizons*, *67*(5), 471–486. https://doi.org/10.1016/j.bushor.2024.03.001
Harasta, J., Novotná, T., & Savelka, J. (2024). It cannot be right if it was written by AI: On lawyers' preferences of documents perceived as authored by an LLM vs a human. *Artificial Intelligence and Law*. https://doi.org/10.1007/s10506-024-09422-w
Hawkins, W., & Mittelstadt, B. (2023). The ethical ambiguity of AI data enrichment: Measuring gaps in research ethics norms and practices. *Proceedings of the 2023 ACM Conference on Fairness, Accountability, and Transparency*, 261–270. https://doi.org/10.1145/3593013.3593995
Healy, K. (2017). Fuck nuance. *Sociological Theory*, *35*(2), 118–127. https://doi.org/10.1177/0735275117709046
Heersmink, R., de Rooij, B., Clavel Vázquez, M. J., & Colombo, M. (2024). A phenomenology and epistemology of large Language models: Transparency, trust, and trustworthiness. *Ethics and Information Technology*, *26*(3), 41. https://doi.org/10.1007/s10676-024-09777-3
Herbold, S., Hautli-Janisz, A., Heuer, U., Kikteva, Z., & Trautsch, A. (2023). A large-scale comparison of human-written versus ChatGPT-generated essays. *Scientific Reports*, *13*(1), 18617. https://doi.org/10.1038/s41598-023-45644-9
Hicks, M. T., Humphries, J., & Slater, J. (2024). ChatGPT is bullshit. *Ethics and Information Technology*, *26*(2), 38. https://doi.org/10.1007/s10676-024-09775-5
Hoel, E. (2024, March 29). A.I.-Generated Garbage Is Polluting Our Culture. *The New York Times*, https://www.nytimes.com/2024/03/29/opinion/ai-internet-x-youtube.html
Jain, S., Suriyakumar, V., Creel, K., & Wilson, A. (2024). Algorithmic pluralism: A structural approach to equal opportunity. *The 2024 ACM Conference on Fairness Accountability and Transparency*, 197–206. https://doi.org/10.1145/3630106.3658899
Jelinek, E. (1995). *Sturm und Zwang: Schreiben als Geschlechterkampf*. Hamburg. Ingrid Klein 1995.
Ji, Z., Lee, N., Frieske, R., Yu, T., Su, D., Xu, Y., Ishii, E., Bang, Y. J., Madotto, A., & Fung, P. (2023). Survey of hallucination in natural Language generation. *ACM Comput Surv*, *55*(12), 248:1–248. https://doi.org/10.1145/3571730
Jiang, C., Xu, H., Dong, M., Chen, J., Ye, W., Yan, M., Ye, Q., Zhang, J., Huang, F., & Zhang, S. (2024a). *Hallucination Augmented Contrastive Learning for Multimodal Large Language Model*. 27036–27046. https://openaccess.thecvf.com/content/CVPR2024/html/Jiang_Hallucination_Augmented_Contrastive_Learning_for_Multimodal_Large_Language_Model_CVPR_2024_paper.html
Jiang, F., Kevin, & Hyland, K. (2024b). Does ChatGPT Argue Like Students? Bundles in Argumentative Essays. Applied Linguistics, amae052. https://doi.org/10.1093/applin/amae052
Judkis, M. (2024, June 30). The deluge of bonkers AI art is literally surreal. *The Washington Post*. https://www.washingtonpost.com/style/of-interest/2024/06/30/ai-art-facebook-slop-artificial-intelligence/
Kalberg, S. (1980). Max weber's types of rationality: Cornerstones for the analysis of rationalization processes in history. *American Journal of Sociology*, *85*(5), 1145–1179. https://doi.org/10.1086/227128
Le Bronnec, F., Verine, A., Negrevergne, B., Chevaleyre, Y., & Allauzen, A. (2024). Exploring Precision and Recall to assess the quality and diversity of LLMs. In Ku L.-W., Martins A.F.T., & Srikumar V. (Eds.), *Proc. Annu. Meet. Assoc. Comput Linguist.* (Vol. 1, pp. 11418–11441). Association for Computational Linguistics (ACL); Scopus. https://www.scopus.com/inward/record.uri?eid=2-s2.0-85204425008&partnerID=40&md5=25cc1d8b81796132e978e3f9d334cdfa
Lee, M. H. J. (2025). *Examining the Robustness of Homogeneity Bias to Hyperparameter Adjustments in GPT-4* (No. https://doi.org/10.48550/arXiv.2501.02211
Lee, H. P. (Hank), Sarkar, A., Tankelevitch, L., Drosos, I., Rintel, S., Banks, R., & Wilson, N. (Eds.). (2025, April 1). *The Impact of Generative AI on Critical Thinking: Self-Reported Reductions in Cognitive Effort and Confidence Effects From a Survey of Knowledge Workers*. Proceedings of the ACM CHI Conference on Human Factors in Computing Systems. https://www.microsoft.com/en-us/research/publication/the-impact-of-generative-ai-on-critical-thinking-self-reported-reductions-in-cognitive-effort-and-confidence-effects-from-a-survey-of-knowledge-workers/

Lee, M. H. J., & Jeon, S. (2025). *Homogeneity Bias as Differential Sampling Uncertainty in Language Models* (No. arXiv:2501.19337). arXiv. https://doi.org/10.48550/arXiv.2501.19337

Lee, M. H. J., & Lai, C. K. (2024). *Probability of Differentiation Reveals Brittleness of Homogeneity Bias in GPT-4* (No. arXiv:2407.07329). arXiv. https://doi.org/10.48550/arXiv.2407.07329

Lee, M. H. J., Montgomery, J. M., & Lai, C. K. (2024). Large Language Models Portray Socially Subordinate Groups as More Homogeneous, Consistent with a Bias Observed in Humans. *The 2024 ACM Conference on Fairness, Accountability, and Transparency*, 1321–1340. https://doi.org/10.1145/3630106.3658975

Liu, H., Xue, W., Chen, Y., Chen, D., Zhao, X., Wang, K., Hou, L., Li, R., & Peng, W. (2024a). *A Survey on Hallucination in Large Vision-Language Models* (No. arXiv:2402.00253). arXiv. https://doi.org/10.48550/arXiv.2402.00253

Liu, Q., Zhou, Y., Huang, J., & Li, G. (2024b). When ChatGPT is gone: Creativity reverts and homogeneity persists. *ArXiv*. https://doi.org/10.48550/ArXiv.2401.06816. No. ArXiv:2401.06816.

Mahdawi, A. (2025, January 8). AI-generated 'slop' is slowly killing the Internet, so why is nobody trying to stop it? *The Guardian*. https://www.theguardian.com/global/commentisfree/2025/jan/08/ai-generated-slop-slowly-killing-internet-nobody-trying-to-stop-it

Maleki, N., Padmanabhan, B., & Dutta, K. (2024). AI Hallucinations: A Misnomer Worth Clarifying. *2024 IEEE Conference on Artificial Intelligence (CAI)*, 133–138. https://doi.org/10.1109/CAI59869.2024.00033

Marco, G., Gonzalo, J., Castillo, R., & Girona, M. T. M. (2024). *Pron vs Prompt: Can Large Language Models already Challenge a World-Class Fiction Author at Creative Text Writing?* (No. arXiv:2407.01119). arXiv. https://doi.org/10.48550/arXiv.2407.01119

Marcus, G., & Davis, E. (2020, August 22). *GPT-3, Bloviator: OpenAI's language generator has no idea what it's talking about*. MIT Technology Review. https://www.technologyreview.com/2020/08/22/1007539/gpt3-openai-language-generator-artificial-intelligence-ai-opinion/

Markelius, A., Wright, C., Kuiper, J., Delille, N., & Kuo, Y. T. (2024). The mechanisms of AI hype and its planetary and social costs. *AI and Ethics*. https://doi.org/10.1007/s43681-024-00461-2

McLuhan, M. (1964). *Understanding media: The extensions of man*. McGraw-Hill.

McQuillan, D. (2022). *Resisting AI: An Anti-fascist approach to artificial intelligence*. Policy.

McQuillan, D. (2023, February 10). *ChatGPT: The world's largest bullshit machine*. Transforming Society. https://www.transformingsociety.co.uk/2023/02/10/chatgpt-the-worlds-largest-bullshit-machine/

Milmo, D. (2023, June 23). Two US lawyers fined for submitting fake court citations from ChatGPT. *The Guardian*. https://www.theguardian.com/technology/2023/jun/23/two-us-lawyers-fined-submitting-fake-court-citations-chatgpt

Moon, K., Green, A., & Kushlev, K. (2024). *Homogenizing Effect of a Large Language Model (LLM) on Creative Diversity: An Empirical Comparison of Human and ChatGPT Writing*. OSF. https://mayukhdifferent.medium.com/why-large-language-models-would-not-draw-a-full-glass-of-wine-87827949d373https://doi.org/10.31234/osf.io/8p9wu

Mukhopadhyay, M. (2025, March 2). Why Large Language Models Would Not Draw a Full Glass of Wine? Medium. https://mayukhdifferent.medium.com/why-large-language-models-would-not-draw-a-full-glass-of-wine-87827949d373

Nguyen, M., Baker, A., Neo, C., Roush, A., Kirsch, A., & Shwartz-Ziv, R. (2024). *Turning Up the Heat: Min-p Sampling for Creative and Coherent LLM Outputs* (No. arXiv:2407.01082). arXiv. https://doi.org/10.48550/arXiv.2407.01082

Okerlund, J., Klasky, E., Middha, A., Kim, S., Rosenfeld, H., Kleinman, M., & Parthasarathy, S. (2022). *WHAT'S IN THE CHATTERBOX? LARGE LANGUAGE MODELS, WHY THEY MATTER, AND WHAT WE SHOULD DO ABOUT THEM*. https://doi.org/10.7302/21898

Oxford Word of the Year 2024. (n.d.). Oxford University Press. Retrieved February 10, (2025). from https://corp.oup.com/word-of-the-year/

Padmakumar, V., & He, H. (2024). *Does Writing with Language Models Reduce Content Diversity?* (No. arXiv:2309.05196). arXiv. https://doi.org/10.48550/arXiv.2309.05196

Palmini, M. T. D. R., & Cetinic, E. (2024). *Patterns of Creativity: How User Input Shapes AI-Generated Visual Diversity* (No. arXiv:2410.06768). arXiv. https://doi.org/10.48550/arXiv.2410.06768

Peeperkorn, M., Kouwenhoven, T., Brown, D., & Jordanous, A. (2024). *Is Temperature the Creativity Parameter of Large Language Models?* (No. arXiv:2405.00492). arXiv. https://doi.org/10.48550/arXiv.2405.00492

Perla, R. J., & Carifio, J. (2007). Psychological, philosophical, and educational criticisms of Harry frankfurt's concept of and views about bullshit in human discourse, discussions, and exchanges. *Interchange*, *38*(2), 119–136. https://doi.org/10.1007/s10780-007-9019-y

Petiska, E. (2023). *ChatGPT cites the most-cited articles and journals, relying solely on Google Scholar's citation counts. As a result, AI May amplify the Matthew effect in environmental science* (No. arXiv:2304.06794). arXiv. https://doi.org/10.48550/arXiv.2304.06794

Porter, B., & Machery, E. (2024). AI-generated poetry is indistinguishable from human-written poetry and is rated more favorably. *Scientific Reports*, *14*(1), 26133. https://doi.org/10.1038/s41598-024-76900-1

Pragya, A. (2024). Generative AI and epistemic diversity of its inputs and outputs: Call for further scrutiny. *AI and Society Scopus*. https://doi.org/10.1007/s00146-024-02097-6

Ragot, M., Martin, N., & Cojean, S. (2020). AI-generated vs. Human Artworks. A Perception Bias Towards Artificial Intelligence? *Extended Abstracts of the 2020 CHI Conference on Human Factors in Computing Systems*, 1–10. https://doi.org/10.1145/3334480.3382892

Rastogi, C., Zhang, Y., Wei, D., Varshney, K. R., Dhurandhar, A., & Tomsett, R. (2022). Deciding Fast and Slow: The Role of Cognitive Biases in AI-assisted Decision-making. *Proc. ACM Hum.-Comput. Interact.*, *6*(CSCW1), 83:1–83:22. https://doi.org/10.1145/3512930

Reif, E., Kahng, M., & Petridis, S. (2023). Visualizing Linguistic Diversity of Text Datasets Synthesized by Large Language Models. *Proc. - IEEE Vis. Conf. - Short Pap., VIS*, 236–240. Scopus. https://doi.org/10.1109/VIS54172.2023.00056

Reviriego, P., Conde, J., Merino-Gómez, E., Martínez, G., & Hernández, J. (2024). Playing with words: Comparing the vocabulary and lexical diversity of ChatGPT and humans. *MACHINE LEARNING WITH APPLICATIONS*, *18*. https://doi.org/10.1016/j.mlwa.2024.100602

Richardson, J. How ChatGPT Fixed the Wine Glass Problem., & Medium (2025, March 30). https://medium.com/@joe.richardson.iii/how-chatgpt-fixed-the-wine-glass-problem-6e75c4851293

Romero, A. (2023, July 12). *ChatGPT: A Bullshit Tool For Bullshit Jobs*. https://www.thealgorithmicbridge.com/p/chatgpt-a-bullshit-tool-for-bullshit

Roselli, D., Matthews, J., & Talagala, N. (2019). Managing Bias in AI. *Companion Proceedings of The 2019 World Wide Web Conference*, 539–544. https://doi.org/10.1145/3308560.3317590

Roth, E. (2023, December 29). *Former Trump lawyer Michael Cohen accidentally cited fake court cases generated by AI*. The Verge. h

ttps://www.theverge.com/2023/12/29/24019067/michael-cohen-former-trump-lawyer-google-bard-ai

Routray, S. K., Javali, A., Sharmila, K. P., Jha, M. K., Pappa, M., & Singh, M. (2023). Large Language Models (LLMs): Hypes and Realities. *Int. Conf. Comput. Sci. Emerg. Technol., CSET*. 2023 International Conference on Computer Science and Emerging Technologies, CSET 2023. Scopus. https://doi.org/10.1109/CSET58993.2023.10346621

Rudko, I., Bashirpour Bonab, A., Fedele, M., & Formisano, A. V. (2024). New institutional theory and AI: Toward rethinking of artificial intelligence in organizations. *Journal of Management History*, *31*(2), 261–284. https://doi.org/10.1108/JMH-09-2023-0097

Rueda-Arango, Y. D., Rojas-Velazquez, D., Gorelova, A. V., & Lopez-Rincon, A. (2024). Exploring Human Perception of AI-Generated Artworks. *2024 IEEE International Symposium on Technology and Society (ISTAS)*, 1–6. https://doi.org/10.1109/ISTAS61960.2024.10732054

Runco, M. A. (2025). The discovery and innovation of AI does not qualify as creativity. *Journal of Cognitive Psychology*, *0*(0), 1–10. https://doi.org/10.1080/20445911.2024.2436362

Salvagno, M., Taccone, F. S., & Gerli, A. G. (2023). Artificial intelligence hallucinations. *Critical Care*, *27*(1), 180. https://doi.org/10.1186/s13054-023-04473-y

Sang, L., Dobolyi, D., & Larsen, K. R. (2024). Assessing Public Sentiment on AI-Generated Art Across the United States Using Deep Learning. *Academy of Management Proceedings*, *2024*(1), 19516. https://doi.org/10.5465/AMPROC.2024.19516abstract

Sastre, A., Iglesias, A., Morato, J., & Sanchez-Cuadrado, S. (2024). Is ChatGPT Able to Generate Texts that Are Easy to Understand and Read? In Á. Rocha, H. Adeli, G. Dzemyda, F. Moreira, & A. Poniszewska-Marańda (Eds.), *Good Practices and New Perspectives in Information Systems and Technologies* (pp. 138–147). Springer Nature Switzerland. https://doi.org/10.1007/978-3-031-60221-4_14

Shahbazi, N., Lin, Y., Asudeh, A., & Jagadish, H. V. (2023). Representation Bias in data: A survey on identification and resolution techniques. *ACM Comput Surv*, *55*(13s), 293:1–293. https://doi.org/10.1145/3588433

Shumailov, I., Shumaylov, Z., Zhao, Y., Papernot, N., Anderson, R., & Gal, Y. (2024). AI models collapse when trained on recursively generated data. *Nature*, *631*(8022), 755–759. https://doi.org/10.1038/s41586-024-07566-y

Shur-Ofry, M. (2023). *Multiplicity as an AI Governance Principle* (SSRN Scholarly Paper No. 4444354). Social Science Research Network. https://doi.org/10.2139/ssrn.4444354

Shur-Ofry, M., Horowitz-Amsalem, B., Rahamim, A., & Belinkov, Y. (2024). *Growing a Tail: Increasing Output Diversity in Large Language Models* (No. arXiv:2411.02989). arXiv. https://doi.org/10.48550/arXiv.2411.02989

Slater, J., & Humphries, J. (2025). Another reason to call bullshit on AI hallucinations. AI & SOCIETY. https://doi.org/10.1007/s00146-025-02346-2

Smith, A. L., Greaves, F., & Panch, T. (2023). Hallucination or confabulation?? Neuroanatomy as metaphor in large Language models. *PLOS Digital Health*, *2*(11), e0000388. https://doi.org/10.1371/journal.pdig.0000388

Sourati, Z., Karimi-Malekabadi, F., Ozcan, M., McDaniel, C., Ziabari, A., Trager, J., Tak, A., Chen, M., Morstatter, F., & Dehghani, M. (2025). *The Shrinking Landscape of Linguistic Diversity in the Age of Large Language Models* (No. arXiv:2502.11266). arXiv. https://doi.org/10.48550/arXiv.2502.11266

Sui, P., Duede, E., Wu, S., & So, R. J. (2024). *Confabulation: The Surprising Value of Large Language Model Hallucinations* (No. arXiv:2406.04175). arXiv. https://doi.org/10.48550/arXiv.2406.04175

Taylor, J. (2025, February 9). Fake cases, judges' headaches and new limits: Australian courts grapple with lawyers using AI. *The Guardian*. https://www.theguardian.com/law/2025/feb/10/fake-cases-judges-headaches-and-new-limits-australian-courts-grappling-with-lawyers-using-ai-ntwnfb

Thompson, B., Dhaliwal, M. P., Frisch, P., Domhan, T., & Federico, M. (2024). *A Shocking Amount of the Web is Machine Translated: Insights from Multi-Way Parallelism* (No. arXiv:2401.05749). arXiv. https://doi.org/10.48550/arXiv.2401.05749

Thrampoulidis, C. (2024). *Implicit Optimization Bias of Next-Token Prediction in Linear Models* (No. arXiv:2402.18551). arXiv. https://doi.org/10.48550/arXiv.2402.18551

Tonmoy, S. M. T. I., Zaman, S. M. M., Jain, V., Rani, A., Rawte, V., Chadha, A., & Das, A. (2024). *A Comprehensive Survey of Hallucination Mitigation Techniques in Large Language Models* (No. arXiv:2401.01313). arXiv. https://doi.org/10.48550/arXiv.2401.01313

Toups, C., Bommasani, R., Creel, K., Bana, S., Jurafsky, D., & Liang, P. S. (2023). Ecosystem-level analysis of deployed machine learning reveals homogeneous outcomes. *Advances in Neural Information Processing Systems*, *36*, 51178–51201.

Vallor, S. (2024). *The AI mirror: How to reclaim our humanity in an age of machine thinking*. Oxford University Press.

Vanneste, B. S., & Puranam, P. (2024). Artificial intelligence, trust, and perceptions of agency. *Academy of Management Review*. https://doi.org/10.5465/amr.2022.0041. amr.2022.0041.

Wachter, S., Mittelstadt, B., & Russell, C. (2024). Do large Language models have a legal duty to tell the truth? *Royal Society Open Science*, *11*(8), 240197. https://doi.org/10.1098/rsos.240197

Wang, A., & Russakovsky, O. (2023). *Overwriting Pretrained Bias with Fine-tuning Data* (No. arXiv:2303.06167). arXiv. https://doi.org/10.48550/arXiv.2303.06167

Wang, J., Zhou, Y., Xu, G., Shi, P., Zhao, C., Xu, H., Ye, Q., Yan, M., Zhang, J., Zhu, J., Sang, J., & Tang, H. (2023). *Evaluation and Analysis of Hallucination in Large Vision-Language Models* (No. arXiv:2308.15126). arXiv. https://doi.org/10.48550/arXiv.2308.15126

Wasielewski, A. (2024). Unnatural images: On AI-Generated photographs. *Critical Inquiry*, *51*(1), 1–29. https://doi.org/10.1086/731729

Wendler, C., Veselovsky, V., Monea, G., & West, R. (2024). Do Llamas Work in English? On the Latent Language of Multilingual Transformers. In L.-W. Ku, A. Martins, & V. Srikumar (Eds.), *Proceedings of the 62nd Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)* (pp. 15366–15394). Association for Computational Linguistics. https://doi.org/10.18653/v1/2024.acl-long.820

Wenger, E. (2024). AI produces gibberish when trained on too much AI-generated data. *Nature*, *631*(8022), 742–743. https://doi.org/10.1038/d41586-024-02355-z

Wolfe, R., & Caliskan, A. (2021). *Low Frequency Names Exhibit Bias and Overfitting in Contextualizing Language Models* (No. arXiv:2110.00672). arXiv. https://doi.org/10.48550/arXiv.2110.00672

Wu, H., & Flanagan, T. (2023). The limits of AI content detectors. *Journal of Student Research*, *12*(3). https://doi.org/10.47611/jsrhs.v12i3.5064

Wu, F., Black, E., & Chandrasekaran, V. (2024). *Generative Monoculture in Large Language Models* (No. arXiv:2407.02209). arXiv. https://doi.org/10.48550/arXiv.2407.02209

Xu, Z., Jain, S., & Kankanhalli, M. (2024). *Hallucination is Inevitable: An Innate Limitation of Large Language Models* (No. arXiv:2401.11817). arXiv. https://doi.org/10.48550/arXiv.2401.11817

Ye, H., Liu, T., Zhang, A., Hua, W., & Jia, W. (2023). *Cognitive Mirage: A Review of Hallucinations in Large Language Models*

(No. arXiv:2309.06794). arXiv. https://doi.org/10.48550/arXiv.2309.06794

Zayed, A., Mordido, G., Shabanian, S., & Chandar, S. (2024). *Should We Attend More or Less? Modulating Attention for Fairness* (No. arXiv:2305.13088). arXiv. https://doi.org/10.48550/arXiv.2305.13088

Zhang, Y., Li, Y., Cui, L., Cai, D., Liu, L., Fu, T., Huang, X., Zhao, E., Zhang, Y., Chen, Y., Wang, L., Luu, A. T., Bi, W., Shi, F., & Shi, S. (2023). *Siren's Song in the AI Ocean: A Survey on Hallucination in Large Language Models* (No. arXiv:2309.01219). arXiv. https://doi.org/10.48550/arXiv.2309.01219

Zhang, S., Xu, J., & Alvero, A. J. (2024). *Generative AI Meets Open-Ended Survey Responses: Participant Use of AI and Homogenization*. OSF. https://doi.org/10.31235/osf.io/4esdp

Žižek, S. (2014). The poetic Torture-House of Language. *Poetry*, *203*(6), 563–566. https://www.jstor.org/stable/43591384

Žižek, S. (2024). *Christian atheism: How to be a real materialist*. Bloomsbury Academic.

Publisher's note Springer Nature remains neutral with regard to jurisdictional claims in published maps and institutional affiliations.